

MongoDB Practical Work Documentation

1. Introduction to MongoDB

MongoDB is a popular **NoSQL database** that stores data in a flexible, **JSON-like document format (BSON)** instead of traditional rows and columns. It is known for its **scalability**, **high performance**, and **ease of use**, making it ideal for modern web applications, big data, and real-time analytics.

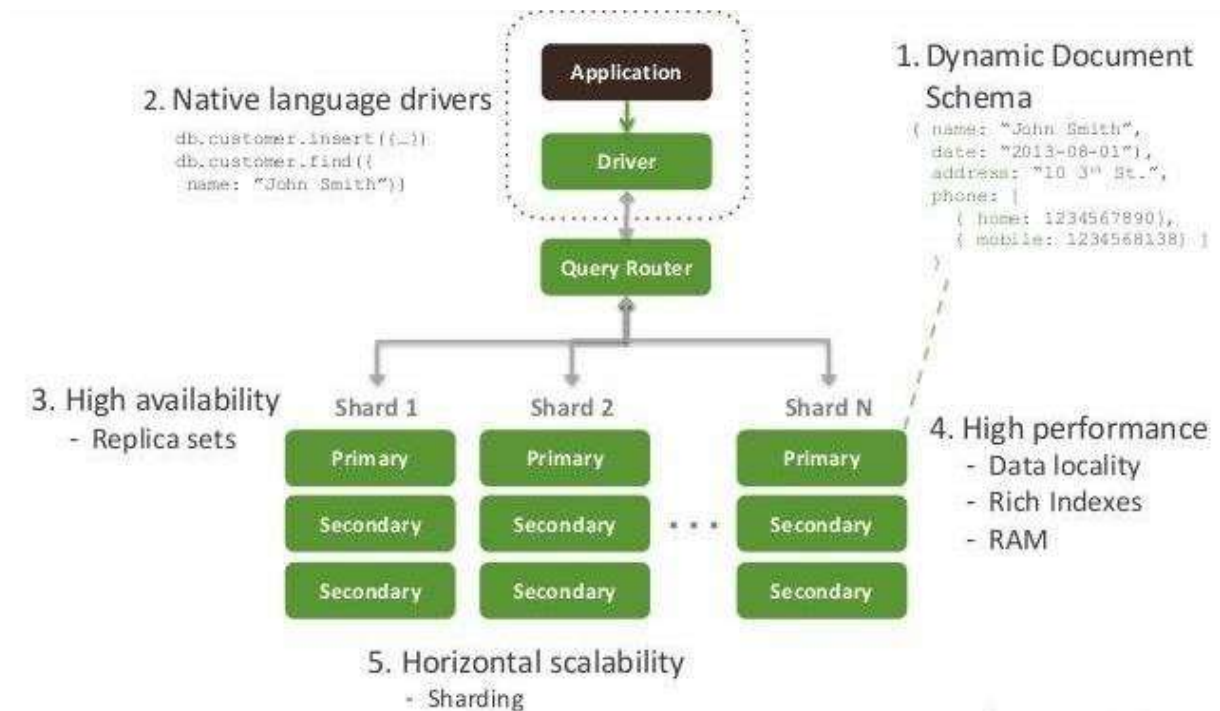
2. MongoDB Architecture

MongoDB follows a **client-server architecture** and consists of the following components:

✿ Key Components:

- **MongoDB Server (mongod)**: Core database engine that handles data requests, manages access, and performs background management operations.
- **Mongo Shell (mongosh)**: Interactive JavaScript interface to interact with MongoDB.
- **Database**: Logical container for collections.
- **Collection**: A group of MongoDB documents (similar to tables).
- **Document**: Basic unit of data in MongoDB (similar to a row in RDBMS), stored as BSON.

Architecture Overview:



3. Installation Steps

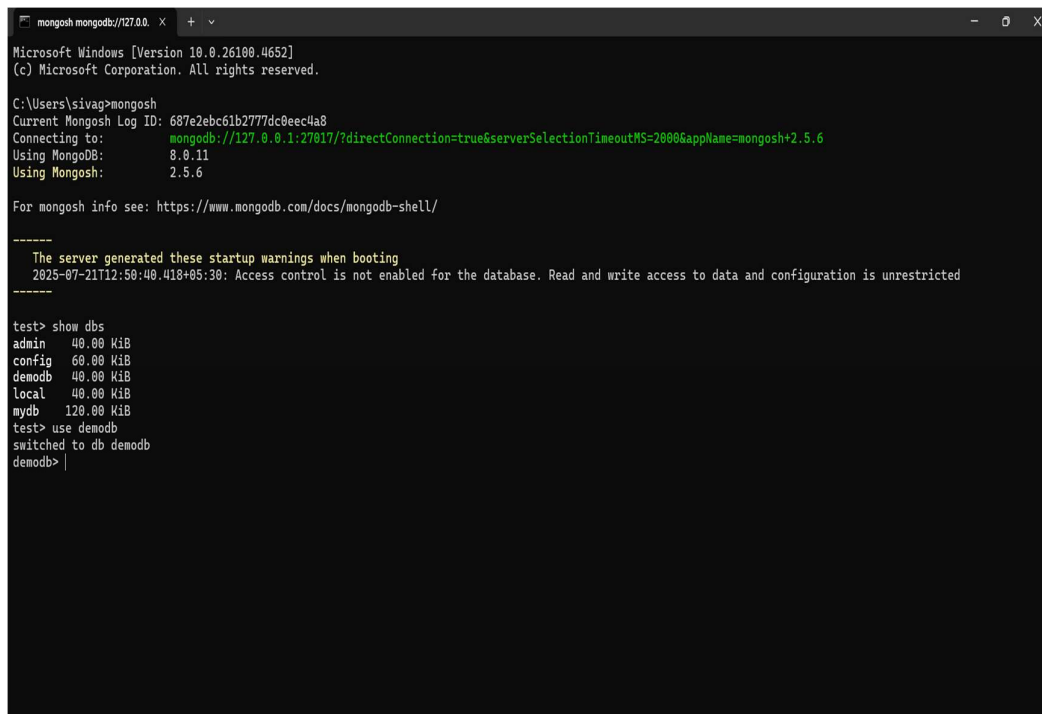
Installed Components:

- MongoDB Community Server (v8.0+)
- Mongo Shell (mongosh)
- MongoDB Compass (optional GUI)

Installation Procedure:

1. Go to the official MongoDB download page: <https://www.mongodb.com/try/download/community>
2. Download the MSI installer for Windows.
3. Run the installer and choose Complete Setup.
4. Enable MongoDB as a Windows Service (recommended).
5. Finish the installation and verify using:

In cmd type - mongosh



```
mongosh mongodbg://127.0.0.1:27020 X + v
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sivag>mongosh
Current Mongosh Log ID: 687e2ebc61b2777dc0eec4a8
Connecting to:  mongodbg://127.0.0.1:27020/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.6
Using MongoDB:  8.0.11
Using Mongosh:  2.5.6

For mongosh info see: https://www.mongodb.com/docs/mongodbg-shell/

-----
The server generated these startup warnings when booting
2025-07-21T12:50:40.418+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
admin    40.00 KiB
config   60.00 KiB
demodb   40.00 KiB
local    40.00 KiB
mydb     120.00 KiB
test> use demodb
switched to db demodb
demodb>
```

4. Executed Queries

Below are the queries executed today in the MongoDB shell (mongosh):

◇ Database & Collection Creation & Record Insertion

```
mydb> use demodb
switched to db demodb
demodb> db.createCollection("people")
{ ok: 1 }
demodb> db.people.insertMany([
...   { user_id: 1, age: 25, status: "passed" },
...   { user_id: 2, age: 30, status: "failed" },
...   { user_id: 3, age: 22, status: "passed" },
...   { user_id: 4, age: 35, status: "failed" },
...   { user_id: 5, age: 28, status: "passed" }
... ])
...
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('687e0e263f06dc6c05eec4b2'),
    '1': ObjectId('687e0e263f06dc6c05eec4b3'),
    '2': ObjectId('687e0e263f06dc6c05eec4b4'),
    '3': ObjectId('687e0e263f06dc6c05eec4b5'),
    '4': ObjectId('687e0e263f06dc6c05eec4b6')
  }
}
```

◆ Fetch Data

```
demodb> db.people.find()
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    age: 25,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    age: 30,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b4'),
    user_id: 3,
    age: 22,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),
    user_id: 4,
    age: 35,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b6'),
    user_id: 5,
    age: 28,
    status: 'passed'
  }
]
```

◆ Fetch Specific Fields

```
demodb> db.people.find({}, { user_id: 1, status: 1 })
```

```
demodb> db.people.find({}, { user_id: 1, status: 1, _id: 0 })
```

```
demodb> db.people.find({}, { user_id: 1, status: 1 })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b4'),
    user_id: 3,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),
    user_id: 4,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b6'),
    user_id: 5,
    status: 'passed'
  }
]
demodb> db.people.find({}, { user_id: 1, status: 1, _id: 0 })
[
  { user_id: 1, status: 'passed' },
  { user_id: 2, status: 'failed' },
  { user_id: 3, status: 'passed' },
  { user_id: 4, status: 'failed' },
  { user_id: 5, status: 'passed' }
]
```

```
demodb> db.people.find({ status: "passed" })
```

```
demodb> db.people.find({ status: "passed" }, { user_id: 1, status: 1, _id: 0 })
```

```
demodb> db.people.find({ status: "passed" })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    age: 25,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b4'),
    user_id: 3,
    age: 22,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b6'),
    user_id: 5,
    age: 28,
    status: 'passed'
  }
]
demodb> db.people.find({ status: "passed" }, { user_id: 1, status: 1, _id: 0 })
[
  { user_id: 1, status: 'passed' },
  { user_id: 3, status: 'passed' },
  { user_id: 5, status: 'passed' }
]
```

```
demodb> db.people.find({ status: { $ne: "passed" } })
```

```
demodb> db.people.find({ status: "passed", age: 25 })
```

```
demodb> db.people.find({ status: { $ne: "passed" } })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    age: 30,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),
    user_id: 4,
    age: 35,
    status: 'failed'
  }
]
demodb> db.people.find({ status: "passed", age: 25 })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    age: 25,
    status: 'passed'
  }
]
```

```
demodb> db.people.find({ $or: [ { status: "passed" }, { age: 25 } ] })
```

```
demodb> db.people.find({ $or: [ { status: "passed" }, { age: 25 } ] })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    age: 25,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b4'),
    user_id: 3,
    age: 22,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b6'),
    user_id: 5,
    age: 28,
    status: 'passed'
  }
]
```

```
demodb> db.people.find({ age: { $gte: 25, $lte: 50 } })
```

```
demodb> db.people.find({ age: { $gte: 25, $lte: 50 } })
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    age: 25,
    status: 'passed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b3'),
    user_id: 2,
    age: 30,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b5'),
    user_id: 4,
    age: 35,
    status: 'failed'
  },
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b6'),
    user_id: 5,
    age: 28,
    status: 'passed'
  }
]
```

```
demodb> db.people.count({ user_id: { $exists: true } })
```

```
demodb> db.people.distinct("status")
```

```
demodb> db.people.count({ user_id: { $exists: true } })
5
demodb> db.people.distinct("status")
[ 'failed', 'passed' ]
demodb> db.people.find().limit(1)
[
  {
    _id: ObjectId('687e0e263f06dc6c05eec4b2'),
    user_id: 1,
    age: 25,
    status: 'passed'
  }
]
```

```
demodb> db.people.find({ status: "A" }).explain()
```

```
demodb> db.people.find({ status: "A" }).explain()
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'demodb.people',
    parsedQuery: { status: { '$eq': 'A' } },
    indexFilterSet: false,
    queryHash: '5D6543D9',
    planCacheShapeHash: '5D6543D9',
    planCacheKey: '405CB45D',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { status: { '$eq': 'A' } },
      direction: 'forward'
    },
    rejectedPlans: []
  },
  queryShapeHash: 'C759020C7D9C8E8E0A2A06FAB1D11A6061F8DD6B59F429B5A09C8C1988863894',
  command: { find: 'people', filter: { status: 'A' }, '$db': 'demodb' },
  serverInfo: {
    host: 'Sivakami',
    port: 27017,
    version: '8.0.11',
    gitVersion: 'bed99f699da6cb2b74262aa6d473446c41476643'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
  }
}
```

5. Conclusion

Today's session helped understand MongoDB's document model and how to install and interact with it using basic queries. I successfully:

- Installed MongoDB.
- Created a database and a collection.
- Inserted documents.
- Queried and filtered data using Mongo Shell.